

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

C++, Python, JavaScript, TypeScript, Java, SQL, Bash, Go (exposure)

Security & Cryptography

AWS KMS envelope encryption (data at rest), authentication, rate limiting, audit logging, multi-tenant isolation, VAPT remediation, secure-by-default design

Backend & Systems

Node.js, Express, FastAPI, Fastify, REST APIs, async services, concurrency, production debugging, integration testing

Cloud & DevOps

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Linux

AI-Assisted Dev

Cursor, Claude / Claude Code, GitHub Copilot, ChatGPT / Codex, design docs, code review

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Built and shipped features to production on a **multi-tenant NHS eTriage platform** for **8+ UK clients** (Node.js, MongoDB) processing **10,000+ monthly submissions**, with a focus on reliability and durability across releases; integrated with Mayden's iaptus platform via REST APIs, contributing to **\$340K+** in revenue.
- Implemented **AWS KMS envelope encryption** to protect PII and clinical data **at rest** and enforced a **90-day data-retention policy** per NHS / client agreement, applying secure-by-default patterns to tenant isolation and backend access paths.
- Owned backend security and reliability across the eTriage stack: hardened **authentication, rate limiting, and audit logging**, added integration tests, monitored production logs during releases, and remediated **20+ Vulnerability Assessment and Penetration Testing (VAPT) findings**.

Cario Growth Services Pvt. Ltd.

November 2022 – May 2023

Backend Developer Intern

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify, backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core codebase from **JavaScript to TypeScript** across multiple repositories and prototyped LLM-powered context-aware replies in a chat product, improving type safety, maintainability, and end-user response quality.

Projects

cf-ai-research-scout - Durable Edge-Native Agent

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Designed **durable workflow boundaries** so multi-step tasks **survive process restarts and individual tool failures**, with agent state and history persisted server-side, prioritizing reliability and durability over a happy-path prototype.
- Built the service end-to-end on Cloudflare Workers with real-time **WebSocket** streaming and embedded storage, taught myself the platform and its concurrency/state model from scratch to ship it.

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, OpenAI / Groq, SQLite, Docker

- Built a multi-stage service that plans typed retrieval, extracts entities with deterministic parsers before LLM fallback, verifies outputs, and returns **cell-level provenance**, treating correctness and auditability as first-class.
- Hardened it with **150+ automated tests**, provider routing/fallback, and pipeline instrumentation, reflecting the same quality and review discipline a platform team depends on.

Sonare - Offline On-Device Speech ↔ Sign App

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

C++ (whisper.cpp), React, Electron, FastAPI, MediaPipe | Qualcomm Edge AI Hackathon 2025

- Built a fully **offline on-device** pipeline (MediaPipe hand tracking, **whisper.cpp** C++ ASR, local TTS) with no external network dependencies, prototyped, designed, and shipped under hackathon time pressure.
- Engineered a **FastAPI** backend and Electron/React frontend with **async APIs and queues** to keep the UI responsive while processing video and audio streams locally on device.

Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.